

Theory Concepts: Regularization, Cross-Validation and Regression Models

Supporting document for the House-Price Regression Pipeline (PR-2).

1. Regularization

Regularization adds a penalty on model complexity (the size of the coefficients) to the training loss. It reduces variance and overfitting and stabilises coefficients when features are correlated or numerous. The strength is controlled by a hyper-parameter (alpha or lambda): 0 = no penalty, very large = strongly simplified model (underfitting).

$$\text{Loss} = \sum (y - \hat{y})^2 + \lambda * \text{Penalty}(w)$$

Trade-off: higher penalty raises bias and lowers variance.

2. Ridge (L2) vs Lasso (L1)

Ridge penalises the sum of squared coefficients. It shrinks coefficients smoothly, keeps all features and copes well with multicollinearity. **Lasso** penalises the sum of absolute coefficients. It can set coefficients exactly to zero, giving automatic feature selection and sparse, interpretable models. Elastic Net combines both.

$$\text{Ridge: } + \lambda * \sum (w_j^2) \quad \text{Lasso: } + \lambda * \sum (|w_j|)$$

Features must be standardised before either penalty is applied, otherwise the penalty depends on the units of each feature.

3. Cross-Validation

Cross-validation estimates how well a model generalises by repeatedly splitting the training data into training and validation parts and averaging the validation scores. It gives a more reliable estimate than one split, and it is used to tune hyper-parameters and compare models without using the test set. Preprocessing (e.g. scaling) must be fitted inside each training fold to avoid data leakage.

4. Cross-Validation Techniques

Technique	How it works	Best used when	Drawback
K-Fold	Split into K folds; each fold is validation once	General default (K = 5 or 10)	Random folds can be unrepresentative
Stratified K-Fold	K-Fold that keeps the class (or binned-target) distribution in every fold	Imbalanced classes or skewed targets	For regression the target must be binned
Leave-One-Out	K = N: one sample is validation, the rest train	Very small datasets	N model fits; high variance
Time Series Split	Expanding window: train on the past, validate on the next block	Time-ordered data, forecasting	Early folds have little training data

5. Tree-based models and feature scaling

A decision tree splits the data with rules of the form 'feature <= threshold'. The best split depends only on the ordering of the values, not their scale, so any monotonic rescaling leaves the tree unchanged. Trees therefore need no standardisation, unlike Ridge, Lasso, SVR or KNN, which use distances, penalties or gradients.

6. Decision Tree and Random Forest

A **Decision Tree Regressor** recursively partitions the feature space and predicts the mean target of each leaf. Deep trees overfit; complexity is controlled with *max_depth*, *min_samples_split* and *min_samples_leaf*. A **Random Forest** averages many trees trained on bootstrap samples with random feature subsets (bagging). Averaging decorrelated trees lowers variance while keeping bias low, and it provides feature importances.

7. Support Vector Regression (SVR)

SVR fits a function that stays within an epsilon-wide tube around the data; errors inside the tube are ignored and only points outside (support vectors) shape the model. Hyper-parameters: **C** (penalty for points outside the tube; large C = less regularisation), **epsilon** (tube width) and, for the RBF kernel, **gamma** (reach of a single training point; large gamma = wiggly, overfit boundary). The linear kernel models straight-line relations; polynomial and RBF kernels capture non-linear ones. SVR is scale-sensitive, so features (and usually the target) must be standardised.

$$K_{\text{rbf}}(x, x') = \exp(-\text{gamma} * ||x - x'||^2)$$

8. Regression metrics

MSE = mean of squared errors (penalises large errors). **RMSE** = $\sqrt{\text{MSE}}$, in the units of the target (INR). **MAE** = mean absolute error, robust to outliers. **R²** = $1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$, the share of variance explained (1 = perfect, 0 = predicting the mean).

9. Bias-variance, overfitting and underfitting

Overfitting: low training error but much higher validation/test error (high variance). **Underfitting**: high error on both training and validation data (high bias). Regularization, limiting tree depth, tuning C/gamma, more data and cross-validation are the main tools to find the balance.